Doppler Effect for Sound Waves (Cambridge (CIE) A Level Physics): Revision Note

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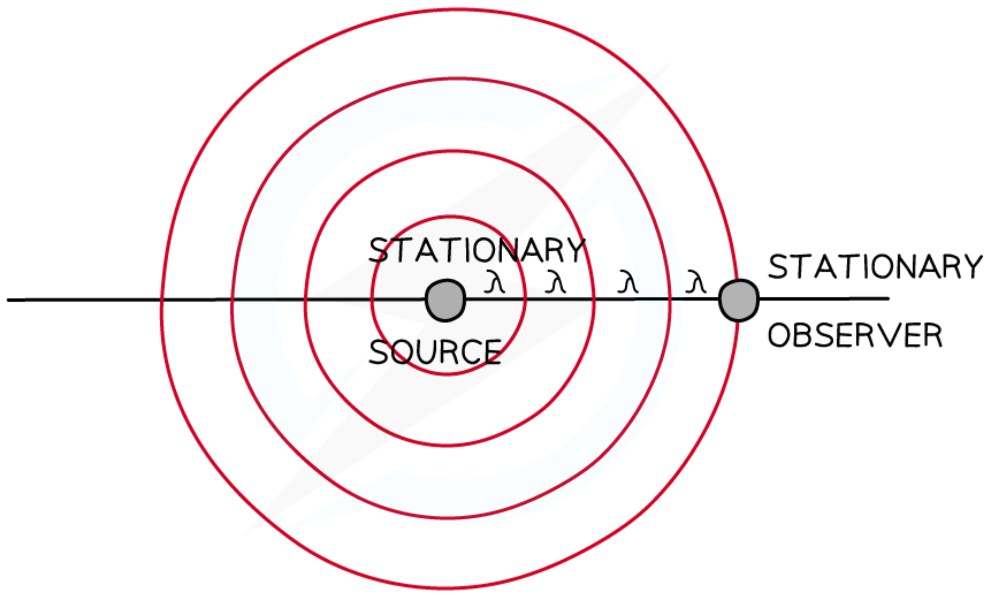
Doppler shift of sound

- The whistle of a train or the siren of an ambulance appears to decrease in frequency (sounds lower in pitch) as it moves further away from you
- This frequency change due to the relative motion between a source of sound or light and an observer is known as the **doppler effect** (or **doppler shift**)
- When the observer (e.g. yourself) and the source of sound (e.g. ambulance siren) are both **stationary**, the waves are at the **same** frequency for both the observer and the source

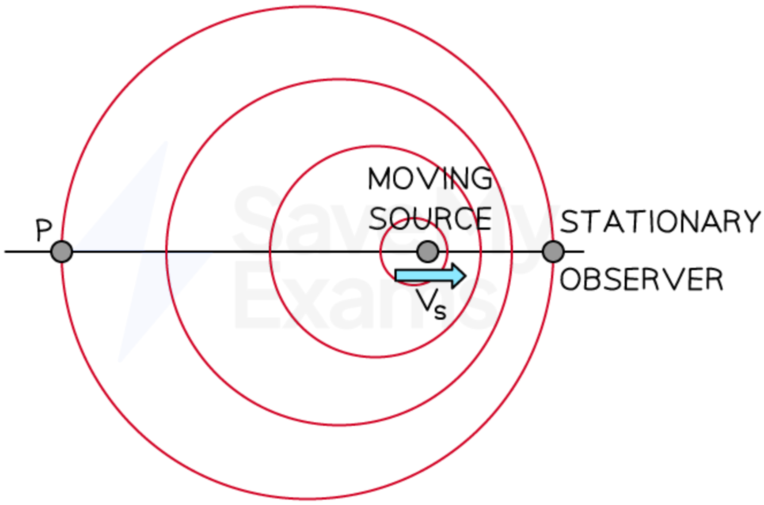
Waves emitted from a stationary source



According to the observer stationary relative to the source, the frequency is equal to that measured by the source

- When the source starts to move **towards** the observer, the wavelength of the waves is **shortened**
  - The sound therefore appears at a **higher** frequency to the observer

Waves emitted by a moving source



The source and observer disagree on their measurements of the frequency of the wave

- Notice how the waves are closer together between the source and the observer compared to point P and the source
- If the observer was at point P instead, they would hear the sound at a lower frequency due to the wavelength of the waves **broadening**
- The frequency is **increased** when the source is moving **towards** the observer
- The frequency is **decreased** when the source is moving **away** from the observer

Worked Example

A cyclist rides past a stationary observer ringing their bell as they ride away.

Which of the following accurately describes the Doppler shift caused by the sound of the bell as the cyclist moves away from the observer?

	Wavelength	**Frequency**	**Pitch**
A	Shorter	Higher	Higher
B	Longer	Lower	Higher
C	Shorter	Higher	Lower
D	Longer	Lower	Lower

Answer: D

- As the cyclist rides away from the observer, the wavelength increases or becomes **longer**
  - This rules out options **A** and **C**
- A longer wavelength means a lower frequency (from the wave equation $v = f\lambda$)
- A lower frequency corresponds to a sound of lower pitch
  - Therefore, the correct answer is **D**

Calculating Doppler shift

- When a **source** of sound waves **moves relative** to a **stationary observer**, the observed frequency can be calculated using the equation below:

$$f_o = f_s\left(\frac{v}{v \pm v_s}\right)$$

- Where
  - f_o is the observed frequency in Hz
  - f_s is the source frequency in Hz
  - v is wave velocity in m s^{-1}
  - v_s is the velocity of the source relative to the observer, again in m s^{-1}

- The wave velocity for sound waves is 340 ms^{-1}
- The ± depends on whether the source is moving towards or away from the observer
  - If the source is moving **towards**, the denominator is $v - v_s$

- If the source is moving **away**, the denominator is $v + v_s$

Worked Example

A police car siren emits a sound wave with a frequency of 450 Hz. The car is travelling away from an observer at speed of 45 m s^{-1}. The speed of sound is 340 m s^{-1}.

Which of the following is the frequency the observer hears?

A. 519 Hz

B. 483 Hz

C. 397 Hz

D. 358 Hz

Answer: C

Step 1: Choose the correct version of the Doppler shift equation:

- The source is moving away from the observer so choose the plus symbol

$$f_o = f_s\left(\frac{v}{v + v_s}\right)$$

Step 2: Substitute the given values:

- The observer's recording of frequency is

$$f_o = 450 \times \left(\frac{340}{340 + 45}\right) = 397 \text{ Hz}$$

Examiner Tips and Tricks

Be careful as to which frequency and velocity you use in the equation. The ‘source’ is always the object which is moving and the ‘observer’ is always stationary.